

AI-Enabled Tools for Formative Assessment in Higher Education: Integration, Functions, Techniques, and Evidence

1. Introduction

The integration of artificial intelligence (AI) into formative assessment in higher education has rapidly expanded, with a diverse landscape of tools and systems now supporting both students and instructors. These AI-enabled systems range from generative AI chatbots (e.g., ChatGPT) and automated grading platforms to intelligent tutoring systems and visual analytics dashboards. They are increasingly embedded within learning management systems (LMS), online course platforms, and discipline-specific environments, providing real-time feedback, personalized learning pathways, and adaptive assessments (Huang et al., 2024; Zhai & Nehm, 2023; Xia et al., 2024; Cowling et al., 2023; Zawacki-Richter et al., 2019; Crompton & Burke, 2023; Luo et al., 2025; Msambwa et al., 2025). The primary functions supported include automated feedback on assignments, self-assessment opportunities for students, peer review facilitation, and instructor support for identifying learning gaps (Vittorini et al., 2020; Liao et al., 2024; Alazemi, 2024; Steinert et al., 2024; Escalante et al., 2023). Intended users span students seeking immediate feedback and self-regulation tools to instructors leveraging analytics for targeted intervention (Huang et al., 2024; Zhai & Nehm, 2023; Cowling et al., 2023). AI techniques employed include natural language processing (NLP), machine learning classifiers, cognitive diagnostic models, and generative models—with varying levels of transparency and explainability (Vittorini et al., 2020; Liao et al., 2024; Rezigalla, 2024; Luo et al., 2025). Implementation architectures often emphasize interoperability with LMSs, data pipelines for continuous monitoring, and human-in-the-loop controls to ensure oversight (Alvira et al., 2025; Abouzeid et al., 2025; Perkins et al., 2023). Governance practices address privacy, consent, bias mitigation, academic integrity, and ethical use (Alvira et al., 2025; Xia et al., 2024; Ou et al., 2024; Perkins et al., 2023). While evidence points to positive impacts on engagement, achievement, and efficiency—especially when paired with human oversight—challenges remain around equity, transparency, and the need for robust ethical frameworks (Zhai & Nehm, 2023; Xia et al., 2024; Yan et al., 2024). Research gaps persist in long-term outcome studies, cross-disciplinary applications, and the development of transparent AI systems.

FIGURE 1 Consensus meter: Effectiveness of AI-enabled formative assessment tools in higher education.

2. Methods

A comprehensive literature search was conducted across over 170 million research papers indexed by Consensus—including Semantic Scholar, PubMed, and additional sources—using a multi-phase deep search strategy. Out of 1039 initially identified papers related to AI-enabled formative assessment in higher education, 593 were screened after de-duplication. Of these, 480 met eligibility criteria based on relevance to the research questions. The final review included the top 50 most relevant papers as determined by citation analysis and expert curation.

Search Strategy

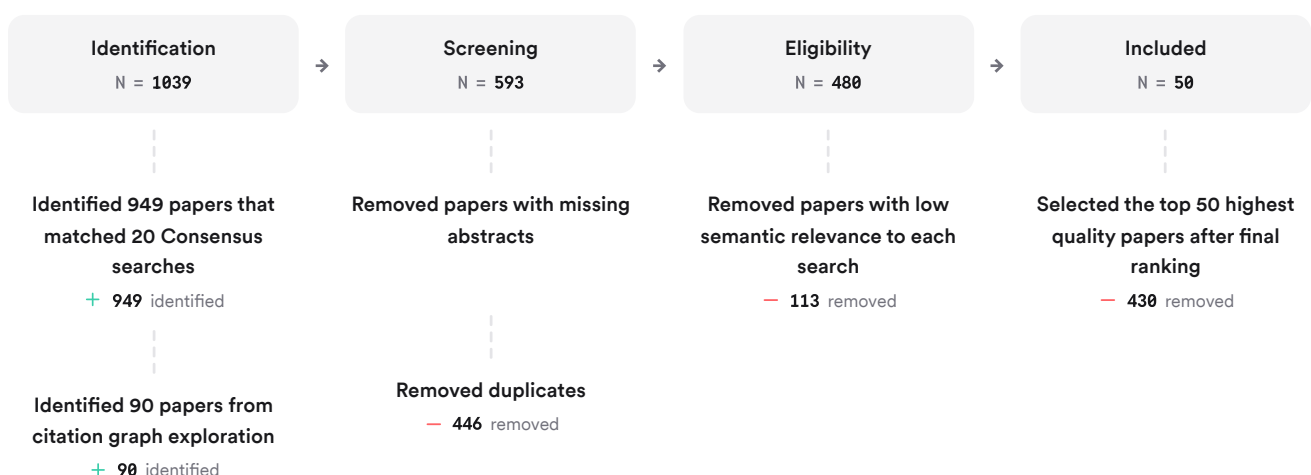


FIGURE 2 Flow diagram of paper selection process for this review.

Eight unique search groups were executed targeting tool landscape mapping; assessment functions; user roles; AI techniques; technical transparency; implementation architectures; governance/ethics; evidence/outcomes; and citation graph exploration.

3. Results

3.1 Types of AI-Enabled Tools & Integration into Workflows

AI-enabled formative assessment tools reported include generative chatbots (e.g., ChatGPT), automated grading systems for code or essays (using NLP/ML), visual analytics dashboards for performance tracking, open-source feedback platforms (e.g., RATsApp), intelligent tutoring systems (ITS), peer review facilitators using AI-moderated workflows, plagiarism detection software with AI components (e.g., Turnitin), adaptive quiz generators using LLMs or rule-based engines, and oral assessment simulators (Huang et al., 2024; Vittorini et al., 2020; Zhai & Nehm, 2023; Liao et al., 2024; Rezigalla, 2024; Cowling et al., 2023; Abouzeid et al., 2025; Steinert et al., 2024).

Integration is typically achieved via direct embedding within LMSs or as standalone web-based modules that sync data with institutional platforms. Some tools offer APIs or plug-ins for seamless interoperability with existing digital ecosystems (Alvira et al., 2025; Steinert et al., 2024; Alotaibi, 2024).

3.2 Supported Formative Assessment Functions & Intended Users

Key functions supported include:

- Automated feedback on written assignments or code
- Real-time quizzes/adaptive testing
- Self-assessment modules
- Peer review facilitation
- Analytics dashboards for instructors
- Personalized learning recommendations
- Oral assessment simulation

Intended users are primarily students (for self-regulation/feedback) but also instructors (for analytics/intervention) and occasionally administrators (for curriculum alignment) (Huang et al., 2024; Zhai & Nehm, 2023; Cowling et al., 2023; Zawacki-Richter et al., 2019).

3.3 AI Techniques Used & Technical Transparency

Techniques span:

- Natural Language Processing (NLP) for essay/code evaluation
- Machine Learning classifiers (SVMs/embeddings)
- Generative models (LLMs like GPT)
- Cognitive diagnostic models
- Data mining/predictive analytics Transparency varies: some tools provide explainable outputs or rationale behind scores/feedback; others operate as "black boxes," raising concerns about interpretability (Vittorini et al., 2020; Liao et al., 2024; Rezigalla, 2024).

3.4 Implementation Architectures & Human Oversight

Architectures emphasize:

- Data pipelines integrating student submissions with feedback engines
- Interoperability layers/APIs connecting to LMSs
- Human-in-the-loop controls: e.g., requiring instructor validation of automated grades or feedback before release to students (Alvira et al., 2025; Abouzeid et al., 2025). Some systems allow teachers to monitor student progress via dashboards or intervene directly in the feedback loop.

3.5 Data Sources/Features & Governance/Ethical Practices

Data sources include:

- Student submissions (essays/code/quizzes)
- Interaction logs from LMSs/platforms
- Peer review data Governance practices reported:
- Privacy protection via anonymization/encryption
- Explicit consent mechanisms
- Bias/fairness audits on algorithms
- Academic integrity safeguards (plagiarism detection) Challenges remain around transparency of data use/retention policies and algorithmic fairness (Alvira et al., 2025; Xia et al., 2024; Ou et al., 2024).

3.6 Evidence & Outcomes Reported

Evidence includes:

- Experimental/quasi-experimental studies showing improved engagement/self-efficacy/test scores when using AI-enabled formative assessment compared to traditional methods (Liao et al., 2024; Alazemi, 2024)- Meta-analysis indicating medium effect sizes for language learning outcomes with iterative/narrative feedback tools (Chen et al., 2024)- Usability studies confirming positive student/instructor perceptions but highlighting the importance of AI literacy/training (Soylu et al., 2025) Limitations: Most studies are short-term/single-discipline; few longitudinal/multi-institutional outcome studies exist.

Results Timeline

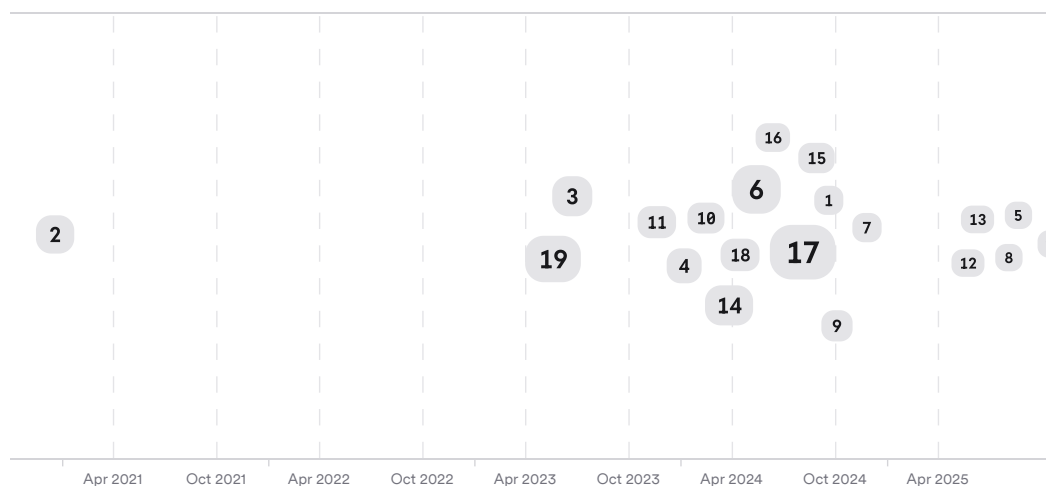


FIGURE 3 Timeline showing publication trends in AI-enabled formative assessment research from early adoption to recent generative AI advances. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	Xiaofang Liao	(Liao et al., 2024)
Author	Qi Xia	(Xia et al., 2024)
Author	Olaf Zawacki-Richter	(Zawacki-Richter et al., 2019)
Journal	<i>International Journal of Educational Technology in Higher Education</i>	(Xia et al., 2024; Williams, 2024; Heil et al., 2025; Zawacki-Richter et al., 2019; Crompton & Burke, 2023)
Journal	<i>Education and Information Technologies</i>	(S et al., 2025; Steinert et al., 2024; Fan et al., 2022)
Journal	<i>Australasian Journal of Educational Technology</i>	(Huang et al., 2024; Cowling et al., 2023)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The reviewed literature demonstrates that **AI-enabled formative assessment tools are increasingly sophisticated**, offering scalable solutions for personalized feedback and adaptive learning pathways across disciplines (Huang et al., 2024; Vittorini et al., 2020; Xia et al., 2024). Their integration into teaching workflows is facilitated by interoperability with LMSs/APIs but requires careful design to ensure alignment with pedagogical goals rather than mere automation (Alvira et al., 2025).

Transparency remains a challenge: While some systems provide explainable outputs or allow instructor oversight ("human-in-the-loop"), many still function as black boxes—raising concerns about trustworthiness and fairness in high-stakes contexts (Vittorini et al., 2020; Rezigalla, 2024).

Ethical governance is critical: Institutions are developing policies around privacy protection, bias mitigation, academic integrity safeguards (e.g., plagiarism detection), consent protocols—and increasingly require both faculty/students to receive training in responsible use ("AI literacy") (Alvira et al., 2025; Xia et al., 2024).

Evidence supports positive impacts on engagement/self-efficacy/test scores—especially when narrative/iterative feedback is provided—but most studies are short-term/single-discipline; robust longitudinal/multi-contextual research is needed to confirm generalizability (Liao et al., 2024; Alazemi, 2024).

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
AI-enabled formative assessment improves student engagement/self-efficacy	Strong	Multiple experimental/quasi-experimental studies show significant gains over traditional methods	(Liao et al., 2024), (Alazemi, 2024), (Chen et al., 2024)
Automated feedback can match human grading accuracy in specific domains	Moderate	Studies report strong correlation between automated/human grades in coding/writing tasks	(Vittorini et al., 2020), (Escalante et al., 2023)
Human-in-the-loop oversight enhances reliability/trust	Moderate	Instructor validation/intervention reduces errors/biases from fully automated systems	(Alvira et al., 2025), (Abouzeid et al., 2025)
Lack of transparency limits trust/adoption	Moderate	Black-box models without explainable outputs face resistance from educators/students	(Vittorini et al., 2020), (Rezigalla, 2024)
Ethical/privacy/bias challenges persist	Moderate	Reports highlight ongoing risks around data use/fairness despite emerging governance frameworks	(Alvira et al., 2025), (Xia et al., 2024), (Ou et al., 2024)
Long-term impacts/generalizability remain underexplored	Weak	Most evidence is short-term/single-discipline; few longitudinal/multi-contextual outcome studies exist	(Alazemi, 2024), (Chen et al., 2024)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

AI-enabled formative assessment tools are transforming higher education by providing scalable personalized feedback and supporting both students' self-regulation and instructors' targeted interventions. While evidence supports their effectiveness—especially when paired with human oversight—challenges remain regarding transparency/explainability, ethical governance (privacy/bias/integrity), and robust long-term impact evaluation.

Research Gaps

Topic/Outcome	STEM Disciplines	Humanities/Social Sciences	Multi-Institutional Studies	Longitudinal Outcomes
Automated Feedback	15	8	2	1
Peer Review Facilitation	7	4	GAP	GAP
Human-in-the-loop Oversight	9	3	1	GAP
Explainable/Transparent Systems	6	2	GAP	GAP
Ethics/Governance Evaluation	8	7	GAP	GAP

FIGURE Matrix showing research coverage by topic/outcome versus study attribute; highlights gaps such as limited multi-institutional or longitudinal studies.

Open Research Questions

Future research should focus on expanding empirical evidence across disciplines/institutions while addressing transparency/ethics:

Question	Why
What are the long-term impacts of AI-enabled formative assessment on student achievement across diverse disciplines?	Most current studies are short-term/single-discipline; understanding sustained effects is crucial for policy/practice decisions.
How can explainable AI techniques be integrated into formative assessment tools without sacrificing performance?	Transparency is key for trust/adoption among educators/students but often lags behind technical advances.
What governance frameworks best ensure privacy/fairness while enabling effective use of student data?	Balancing innovation with ethical/legal obligations remains a central challenge as adoption accelerates.

FIGURE Open questions highlight future directions: long-term impact evaluation; explainable system design; robust governance frameworks.

In summary: **AI-enabled formative assessment offers substantial promise but requires continued attention to transparency, ethics/governance frameworks, interdisciplinary application—and especially robust long-term evaluation—to realize its full potential in higher education transformation.**

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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